筑波大学
University of Tsukuba

TARGET-DEPENDENT CUE RANKINGS IN PROSODIC PROMINENCE: COMPARING HUMAN AND CWT LABELS

The target definition changes the acoustic conclusions we draw.

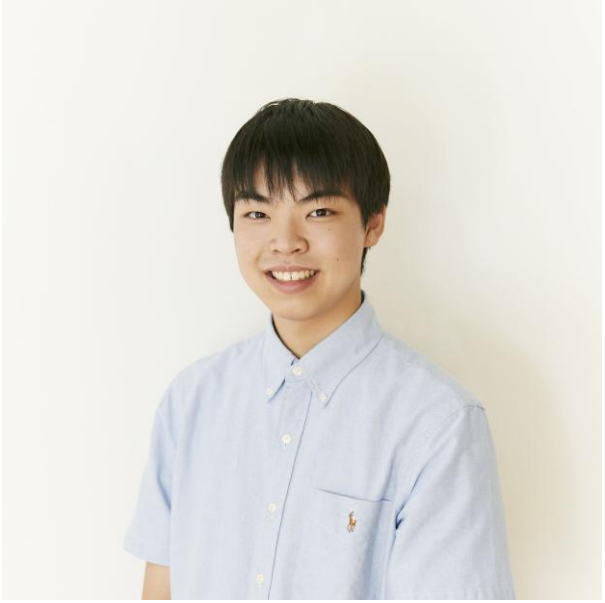
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Self-Introduction



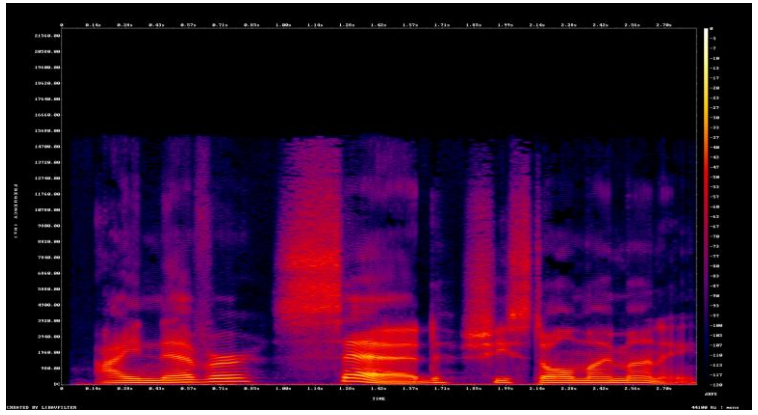


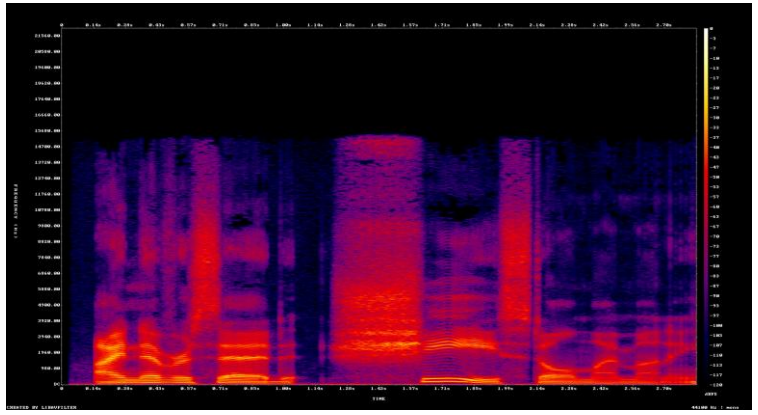
- Kosuke Shimizu
- B3, Collage of Media Arts, Science and Technology,
  The University of Tsukuba
- Researching Human-Computer Interaction
- First time to participate in ICSV conference!

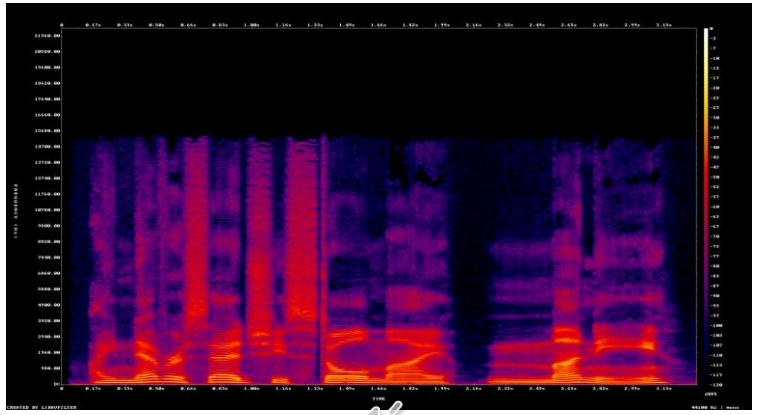
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What is Prosodic Prominence?

- Prominence is the PERCEIVED importance or emphasis of a word in speech.
  - I never SAID she stole my money. (I only implied it.)
  - I never said SHE stole my money. (Someone stole it.)
  - I never said she stole my MONEY. (She stole something.)

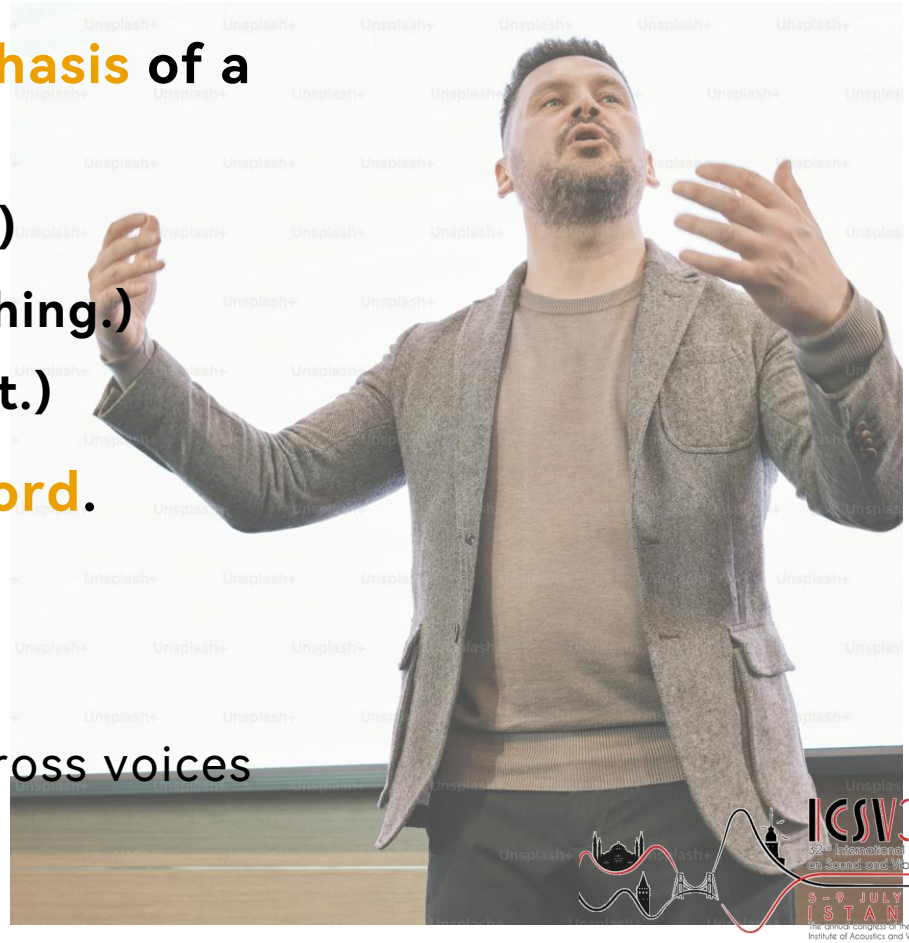






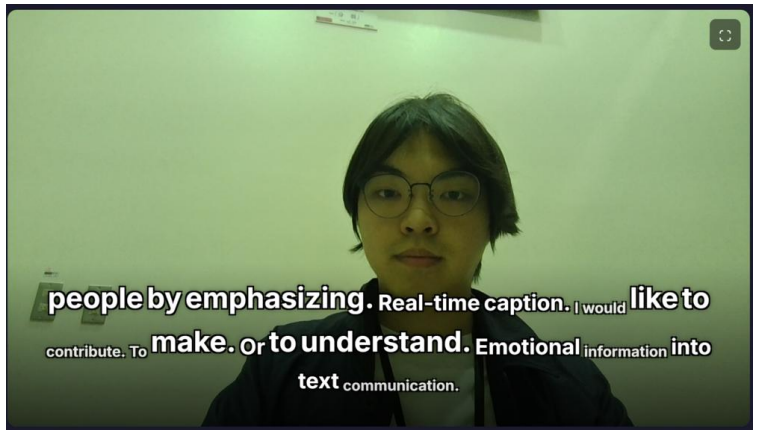
What is Prosodic Prominence?

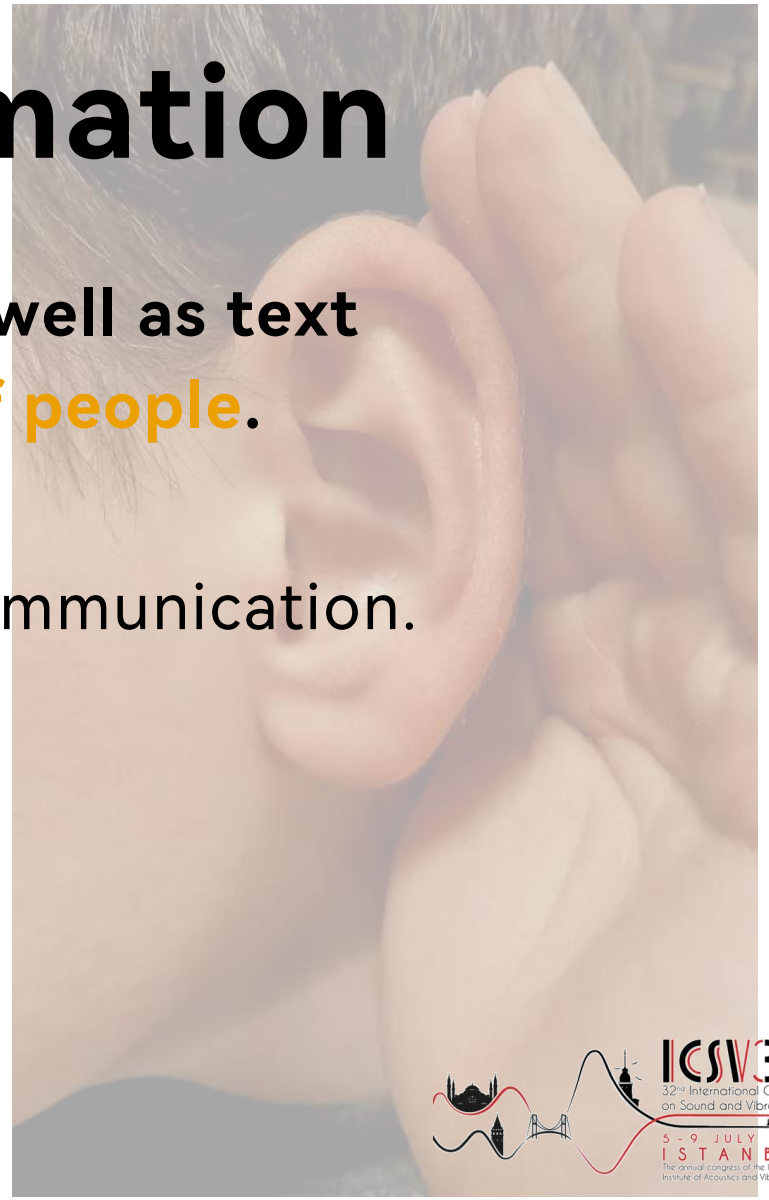
- **Prominence is the PERCEIVED importance or emphasis of a word in speech.**
  - **I never SAID she stole my money. (I only implied it.)**
  - **I never said she stole my MONEY. (She stole something.)**
  - **I never said SHE stole my money. (Someone stole it.)**
- **Same sentence structure, different highlighted word.**
- **Why So Matter**
  - **Text-to-speech:** controlling emphasis and prosody
  - **Speech-to-speech systems:** preserving emphasis across voices or languages



Using Prominence Information

- **Our goal is to convey prosodic information as well as text information in real-time transcription for Deaf people.**
  - i.e.) Augmented Reality Glasses, Realtime Transcription
- System should be fast enough to support live communication.





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Automating Prominence Labeling

Widely used because it is fast, consistent, and scalable when human prominence labels are unavailable.(i.e. Weakly Supervised Learning)

- **[Transform]** Compute multi-scale waveform variations using **CWT**.
- **[Extract]** Isolate **the specific frequency scale** for word-level emphasis.
- **[Detect]** **Identify local amplitude peaks** within the filtered signal.
- **[Binarize]** Align peaks with words and apply a threshold to label.

Prominence is the

PERCEIVED importance or

emphasis of a word in speech.

Is it the same with acoustic-based modeling?
How the difference are between human's and automated labeling method?

What Exactly is the Target?

- **Human-perceived prominence**
  What LISTNERS hear as prominent
- **CWT-derived prominence**
  What an automatic signal-based method marks as prominent
- **Key question:**
  Do these targets support the same acoustic conclusions?
- **and our novelty is severely testing this difference**

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Method & Data

- **Dataset: LibriTTS**(69,714 words from 17 speakers| Read)
  - Human: mean Emphases crowdsourced score, thresholded at 0.5 from Emphase
  - CWT: token-level from automatic evaluation bundle
- **Acoustic Cue Families (F1-F7)**
  - F1: Duration only | F2: + Energy | F3: + F0 | F4/F5: + Cepstral A/B | F7: Full
- **Evaluation Protocol**
  - Speaker-grouped 5-fold cross-validation, logistic regression (balanced weights)
  - Metrics: AUC, AP, Brier, ECE-10 | 3 tasks: Human / CWT / disagreement direction

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Research Design Overview

Same words, same acoustic features, different target definitions

Same Speech Tokens

69,714 words from 17 speakers

Read speech

Human-perceived prominence

Crowdsourced Emphases mean score

Thresholded at 0.5

CWT-derived prominence

Automatic signal-based label

Token-level CWT prominence

Same Acoustic Cue Families

- Duration
- Energy
- F0
- Cepstral proxies

Shared compact acoustic representation

Compare Across Target Definitions

Label agreement | Discrimination | Calibration | Cue ranking | Mismatch direction

Key Question

Do Human and CWT labels lead to the same acoustic conclusions?

Target definition is the experimental factor being tested.

Results: Label Agreement

- **Human vs. CWT: Related but Not Equivalent**

		CWT label	
		CWT negative	**CWT positive**
Human label	**Human negative**	**13,627** 19.55% of tokens	**21,985** 31.54% of tokens
	Human positive	**3,043** 4.36% of tokens	**31,059** 44.55% of tokens

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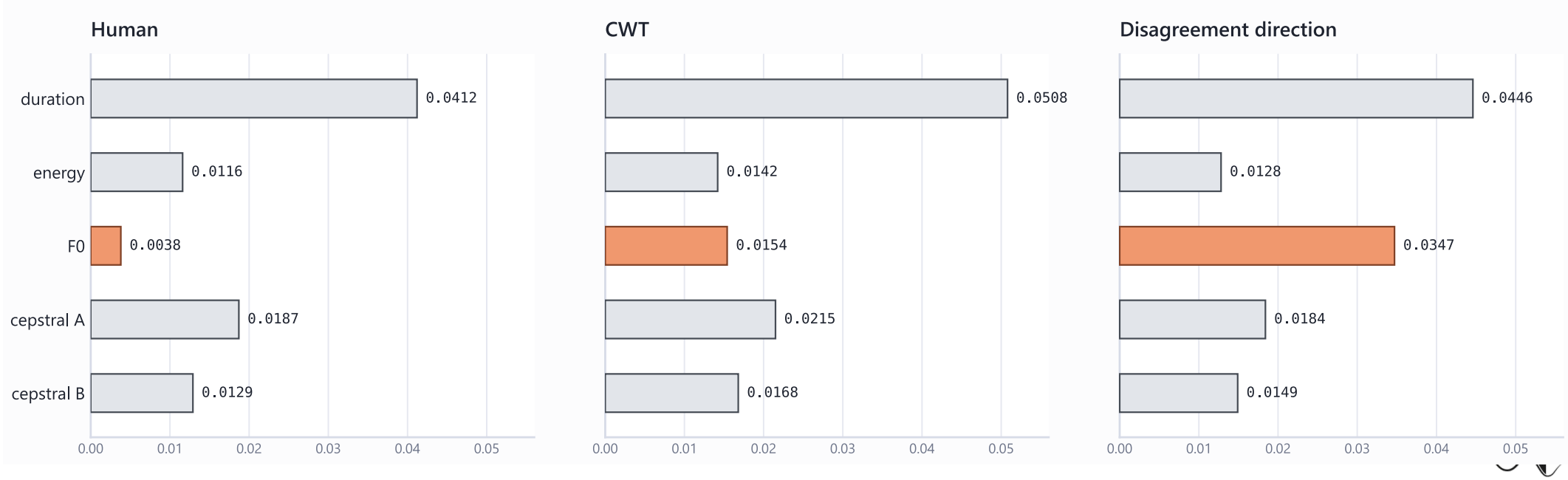
Results: Label Agreement

- **Human vs. CWT: Related but Not Equivalent**
  - Agreement rate: 64% | Cohen's kappa: 0.29 | MCC: 0.34
  - 25,028 mismatched tokens (35.9% of dataset)
  - Most mismatches: CWT-positive but Human-negative (21,985 tokens, ~7:1 ratio)
- **Discrimination (Full Acoustic Family, AUC)**
  - Human target: AUC = 0.744, CWT target: AUC = 0.819
  - Gap = +0.075 (95% CI [0.065, 0.083], bootstrap confirmed)
  - Stable across random forest (+0.071) and n-gram surprisal augmentation (+0.078)

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Cue Rankings Shift with Target

- **Duration: backbone for all tasks**
- **F0 contribution is target-dependent**



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Cue Rankings Shift with Target

- **Duration: backbone for all tasks**
  - Permutation AUC-drop: Human 0.182 | CWT 0.122 | Disagreement 0.188
- **F0 contribution is target-dependent**
  - Human: F0 AUC-drop = 0.0038 (marginal)
  - CWT: F0 AUC-drop = 0.0154 (4x larger)
  - Kochanski et al. (2005): “F0 lends little” -- closer to Human target, not CWT
- **Calibration also differs**
  - ECE-10: Human = 0.215 (overconfident) | CWT = 0.049 (well-calibrated)
  - Same acoustic features produce different probability scaling per target

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Mismatch is not random

- Many mismatches were words that were locally salient in the acoustic signal but were not selected by listeners as emphasis.
  - CWT tends to mark local acoustic peaks.
  - Human labels appear more selective.
- CWT-positive / Human-negative examples:
  coming, happy, madame, advantage, responded, people, must, just, why
- A signal-based target can make prominence look more F0- and peak-driven than human perception does.

Implications & Limitations

- **Scientific Implications**
  - Cue rankings are partly target-contingent, not pure properties of prominence.
  - Cross-study disagreements may reflect unreported target mismatch.
  - Affects downstream TTS, prosody prediction, prominence-conditioned generation.
- **Actionable Recommendation**
  - Report the target definition explicitly -- it is a first-class experimental factor
  - Use matched-token comparisons to separate robust from target-contingent cues
- **Limitations**
  - One corpus, 17 speakers, read speech | MFCC proxies, not full spectral balance

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Conclusion

- **Key Findings**
  - Human and CWT labels agree only 64% -- not interchangeable
  - CWT is more acoustically predictable (AUC 0.819 vs. 0.744)
  - F0 matters **4x** more for CWT -- cue rankings are target-contingent
- **Report target definition explicitly!**
  - Target choice is a first-class experimental factor, not a labeling detail
- **New algorithm for this task @ Interspeech 2026!**



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